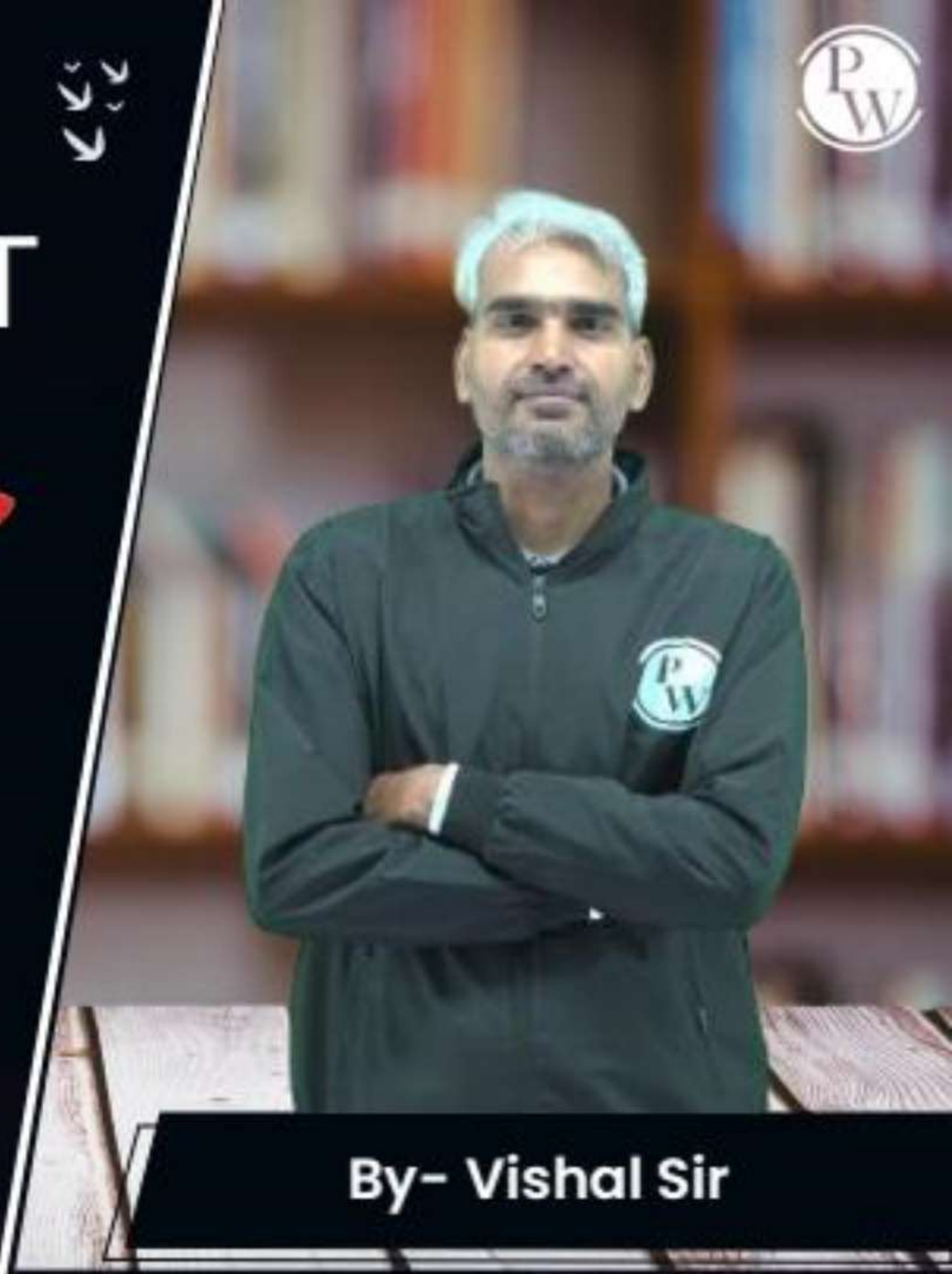
Computer Science & IT



Database Management System

Entity Relationship Model & Integrity constraints

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture





Topic B+ Tree insertion and deletion



Slide

Topics to be Covered

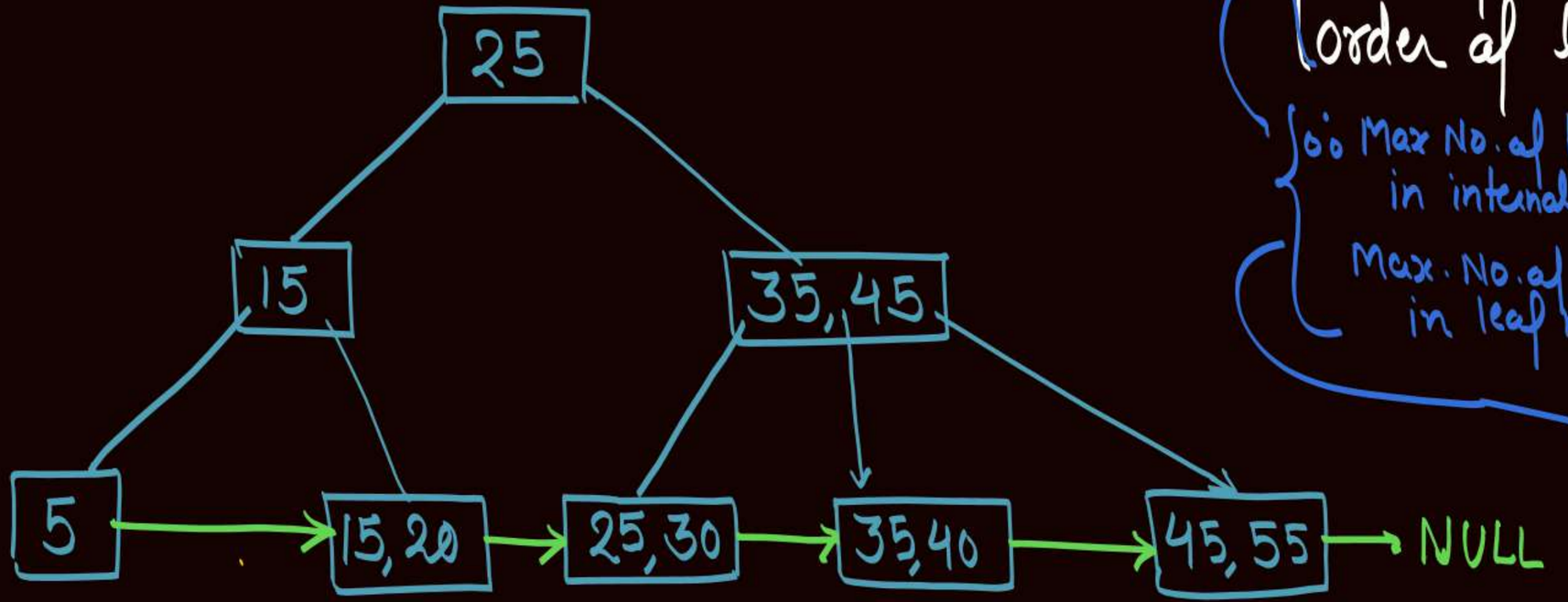


- **Topic** ER model & ER diagram
- **Topic** Relational model & Integrity constraints

Slide

H.W.

Q:- Consider the following B+ tree



Let,
Order of internal node = 3
Order of leaf node = 2

∴ Max No. of keys in internal node = 3-1 = 2

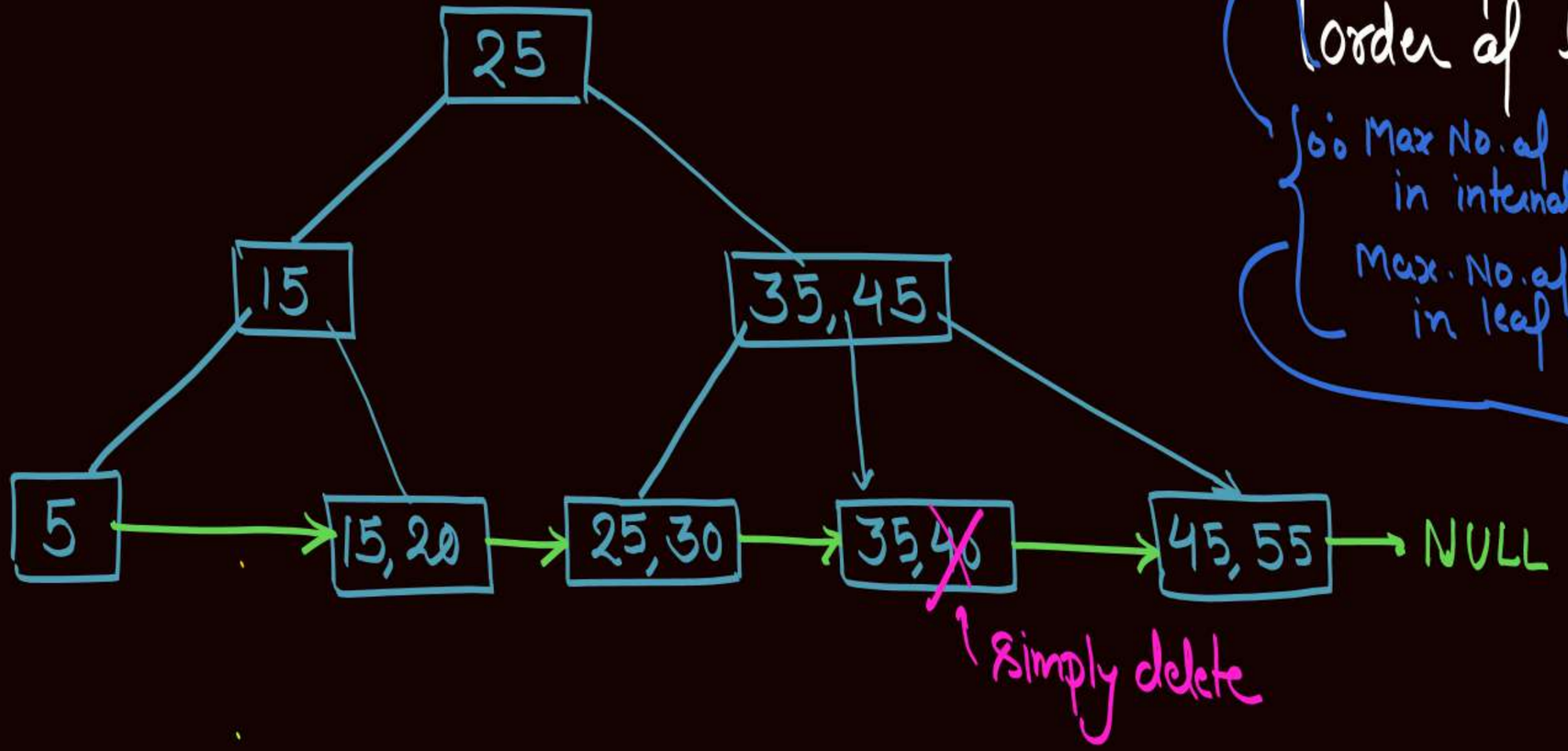
Max. No. of keys in leaf node = 2

Min keys in internal node: $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node: $\lceil \frac{2}{2} \rceil = 1$

Delete the following elements from the above B+ tree in the same order 40, 5, 45, 35, 25, 55

delete ⇒ ~~40~~, 5, 45, 35, 25, 55



Let,
Order of internal node = 3
Order of leaf node = 2

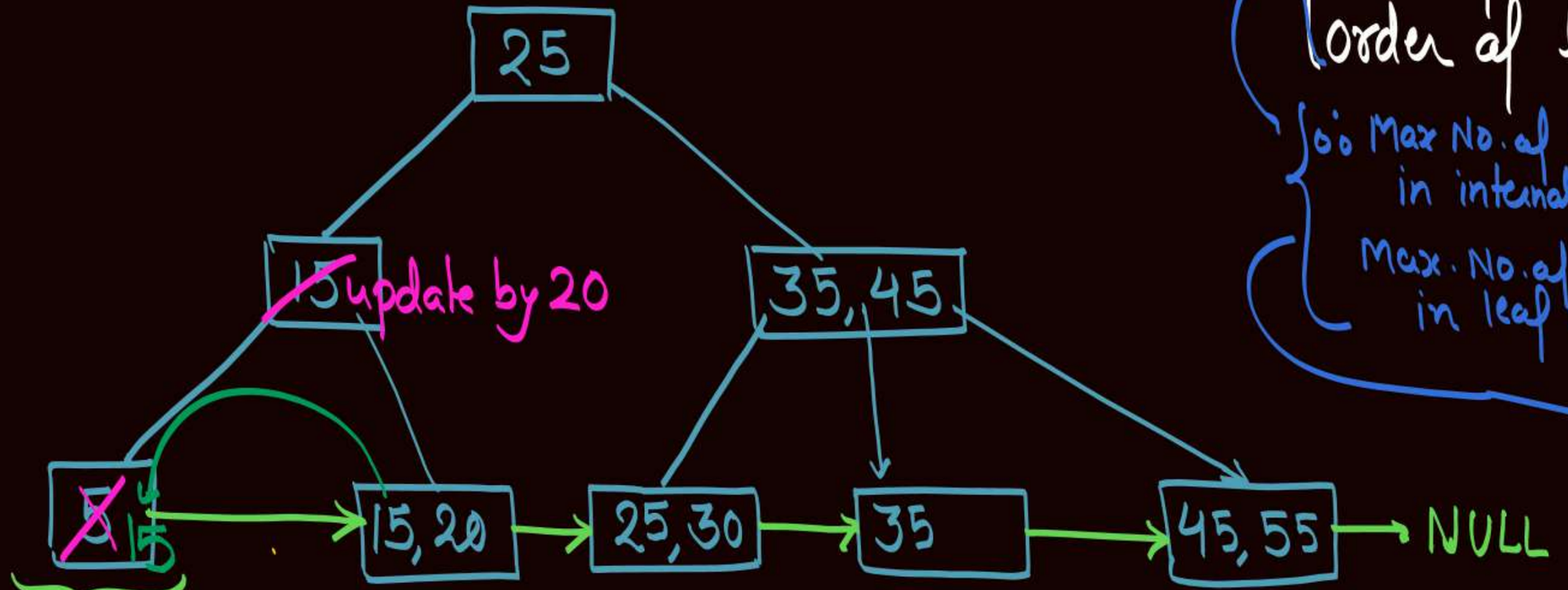
∴ Max No. of keys in internal node = 3 - 1 = 2

Max. No. of keys in leaf node = 2

Min keys in internal node : $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, 45, 35, 25, 55



Present only at leaf

Underflow

∴ ask siblings { Right sibling can help
∴ Redistribute from Right sibling

Let,
Order of internal node = 3
Order of leaf node = 2

∴ Max No. of keys in internal node = 3-1 = 2

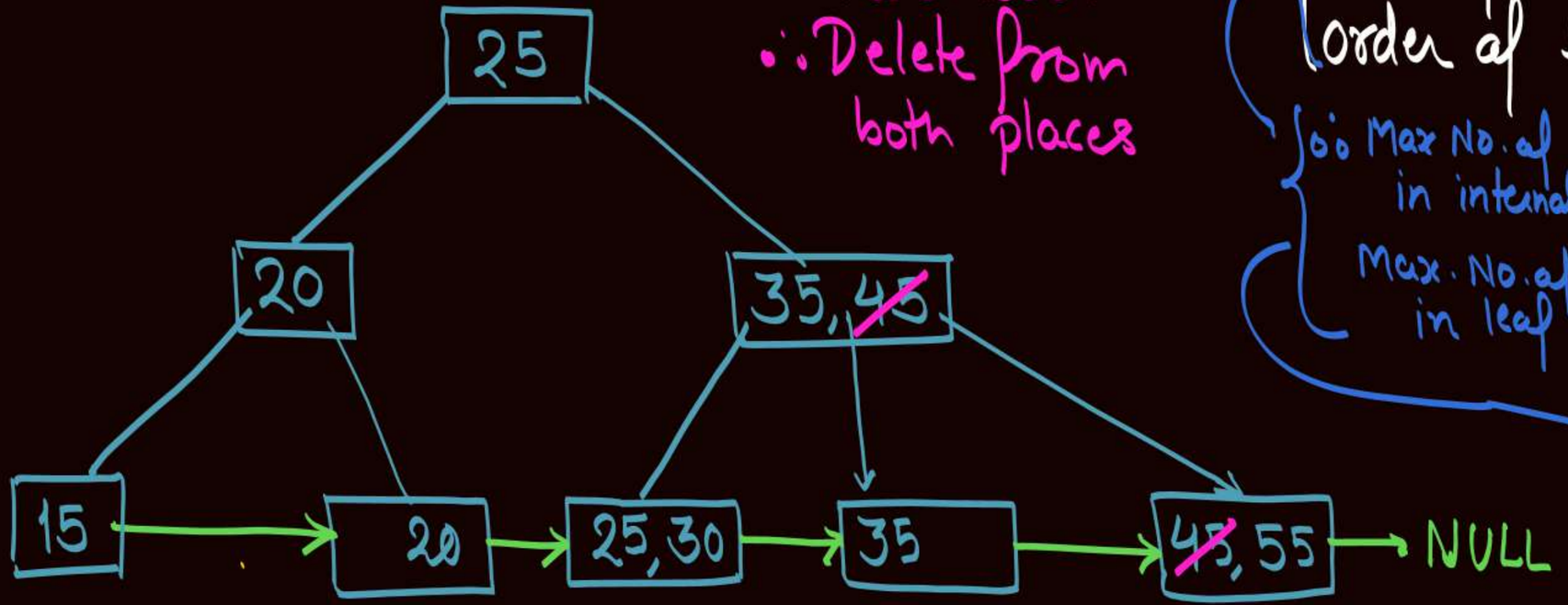
Max. No. of keys in leaf node = 2

Min keys in internal node : $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, ~~45~~, 35, 25, 55

↳ Present in leaf node as well as in internal node
∴ Delete from both places



Let,
Order of internal node = 3
Order of leaf node = 2

∴ Max No. of keys in internal node = 3-1 = 2

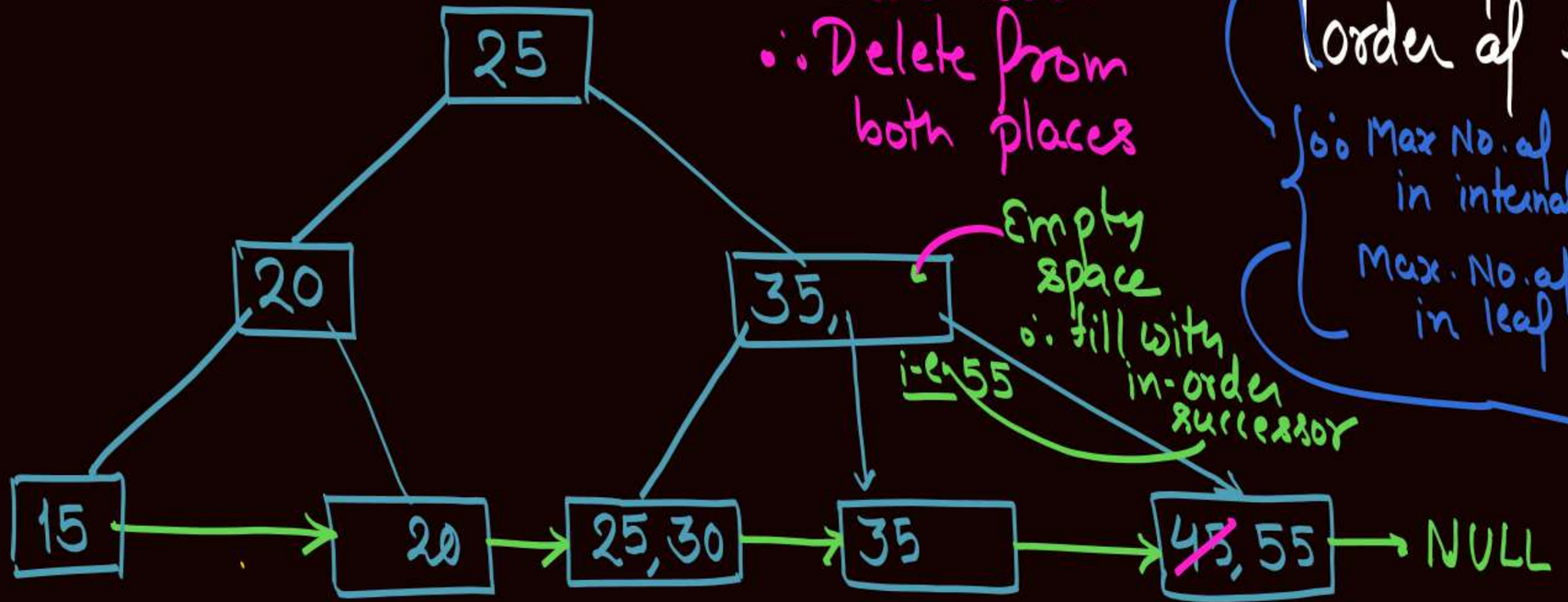
Max. No. of keys in leaf node = 2

Min keys in internal node: $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node: $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, ~~45~~, 35, 25, 55

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∴ Delete from both places



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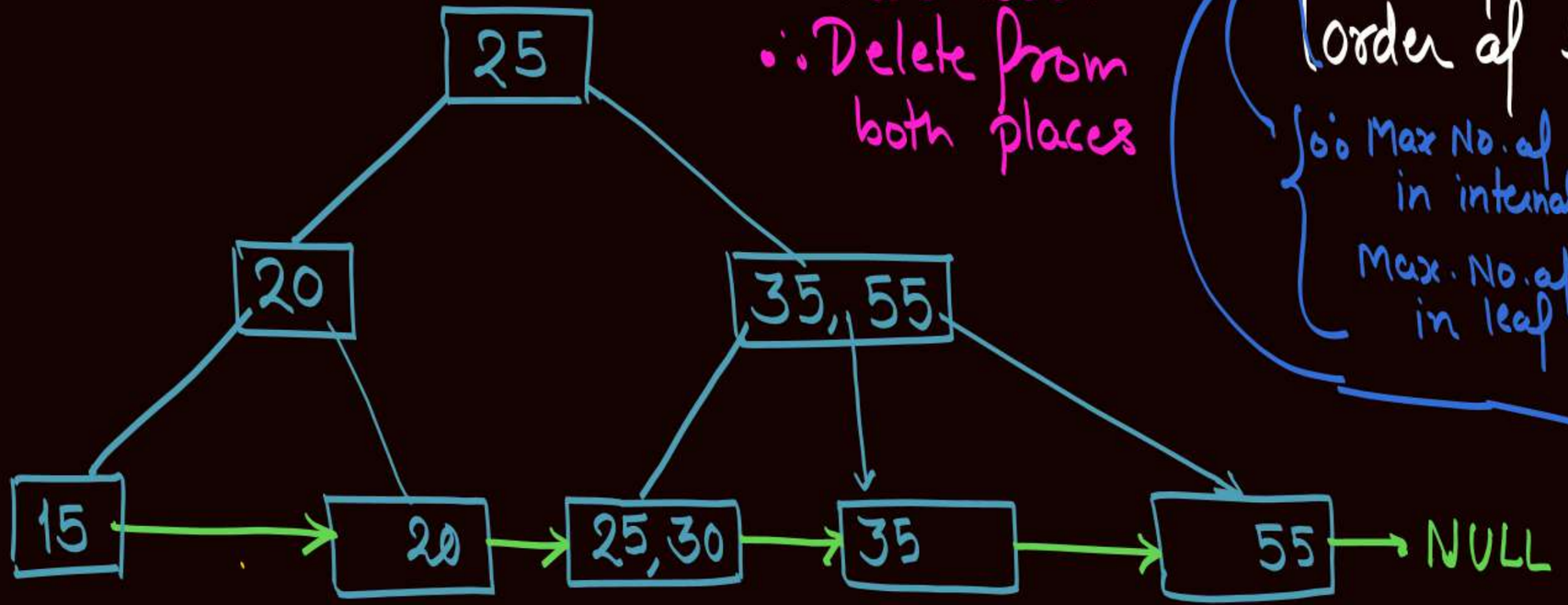
Max. No. of keys in leaf node = 2

Min keys in internal node : $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

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∴ Delete from both places



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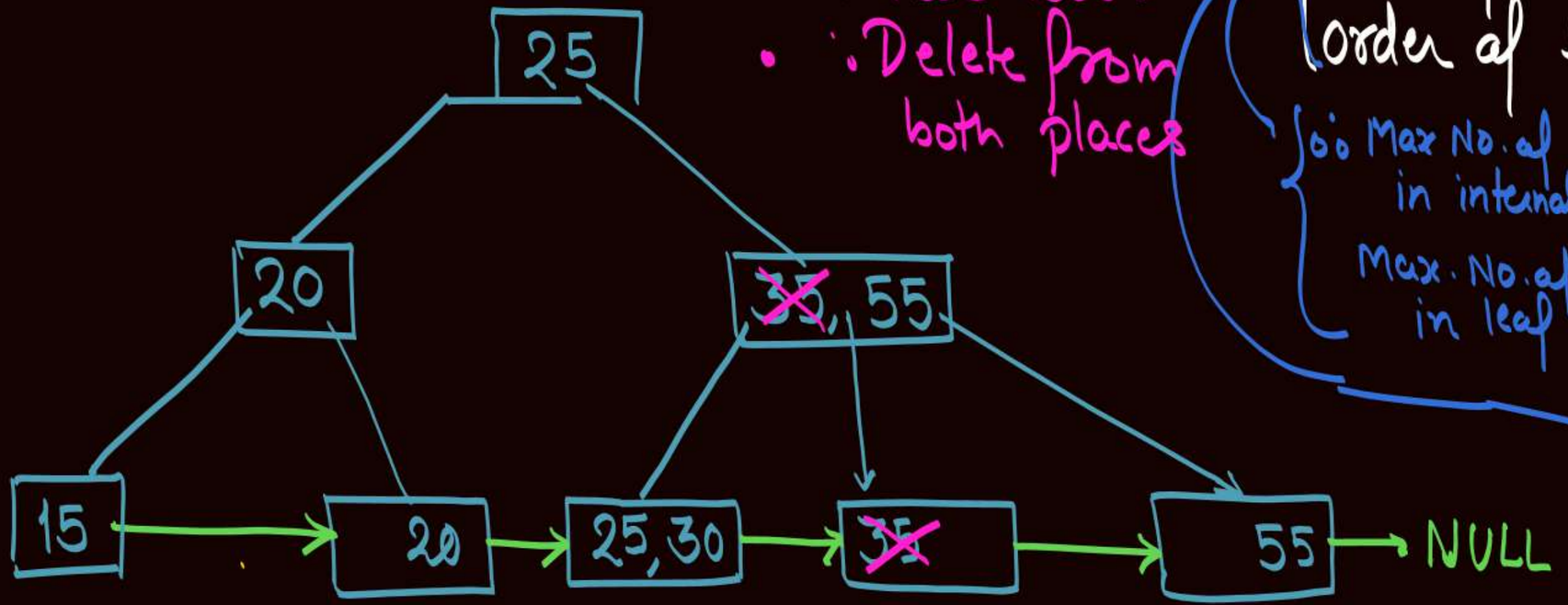
Max. No. of keys in leaf node = 2

Min keys in internal node : $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, 25, 55

Present in leaf node as well as in internal node
∴ Delete from both places



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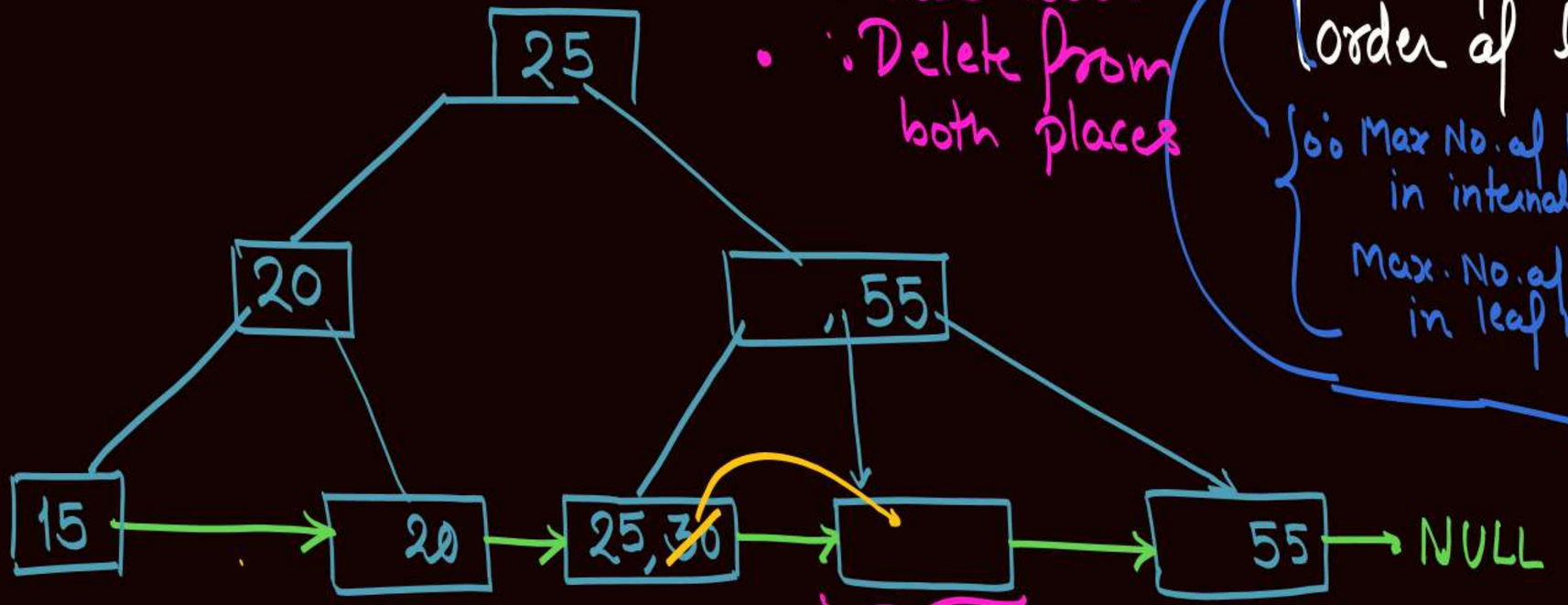
Max. No. of keys in leaf node = 2

Min keys in internal node : $\lceil \frac{3}{2} \rceil - 1 = 1$

Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, 25, 55

Present in leaf node as well as in internal node
∴ Delete from both places



Let,
Order of internal node = 3
Order of leaf node = 2

∴ Max No. of keys in internal node = 3-1 = 2
Max. No. of keys in leaf node = 2

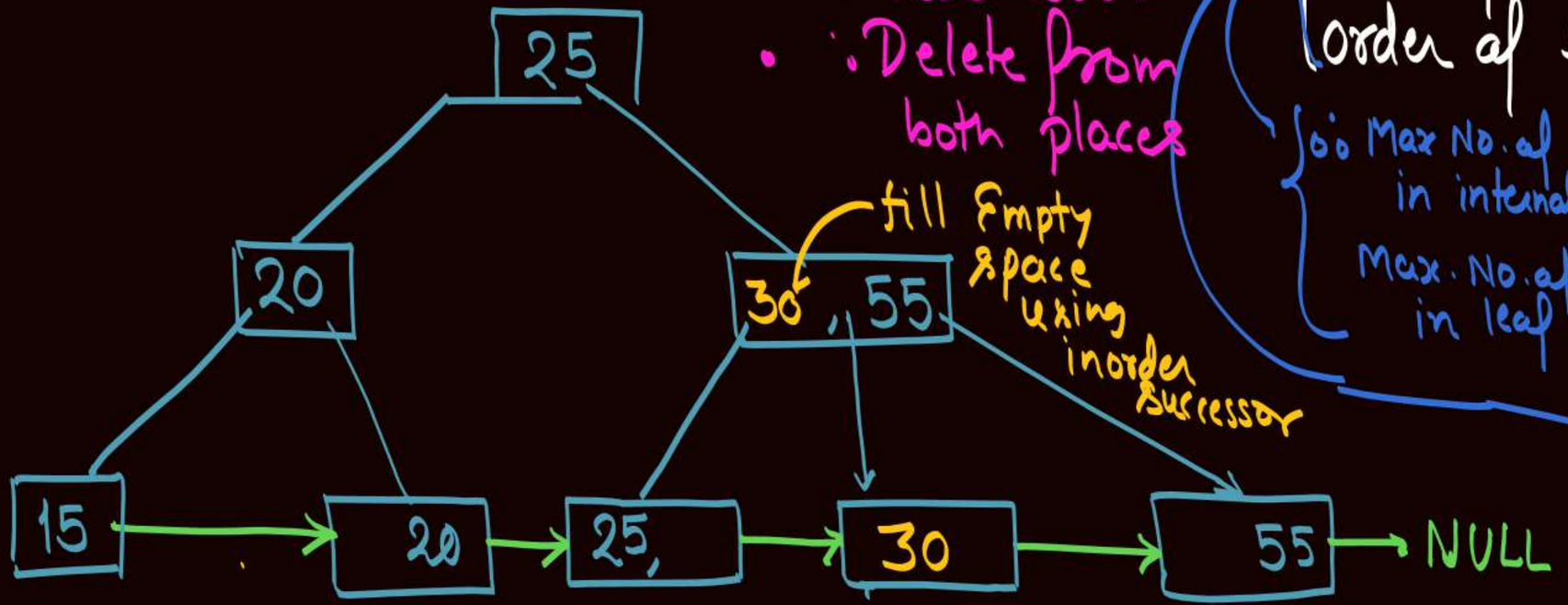
Min keys in internal node: $\lceil \frac{3}{2} \rceil - 1 = 1$
Min No. of keys in leaf node: $\lceil \frac{2}{2} \rceil = 1$

underflow
∴ Borrow from sibling {if possible}
left sibling can help. ∴ Borrow from left sibling

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, 25, 55

Present in leaf node as well as in internal node

∴ Delete from both places



Let,
Order of internal node = 3
Order of leaf node = 2

∴ Max No. of keys in internal node = 3-1 = 2

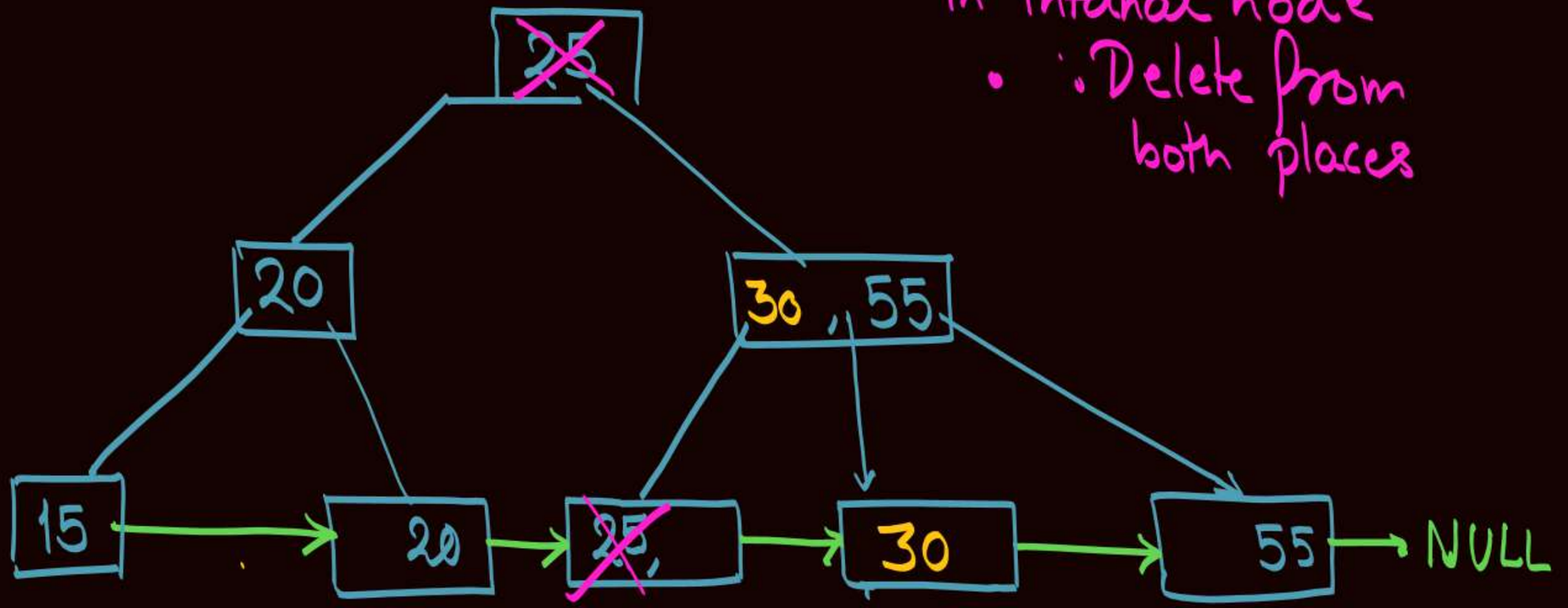
Max. No. of keys in leaf node = 2

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Min No. of keys in leaf node : $\lceil \frac{2}{2} \rceil = 1$

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

Present in leaf node as well as in internal node
∴ Delete from both places



delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

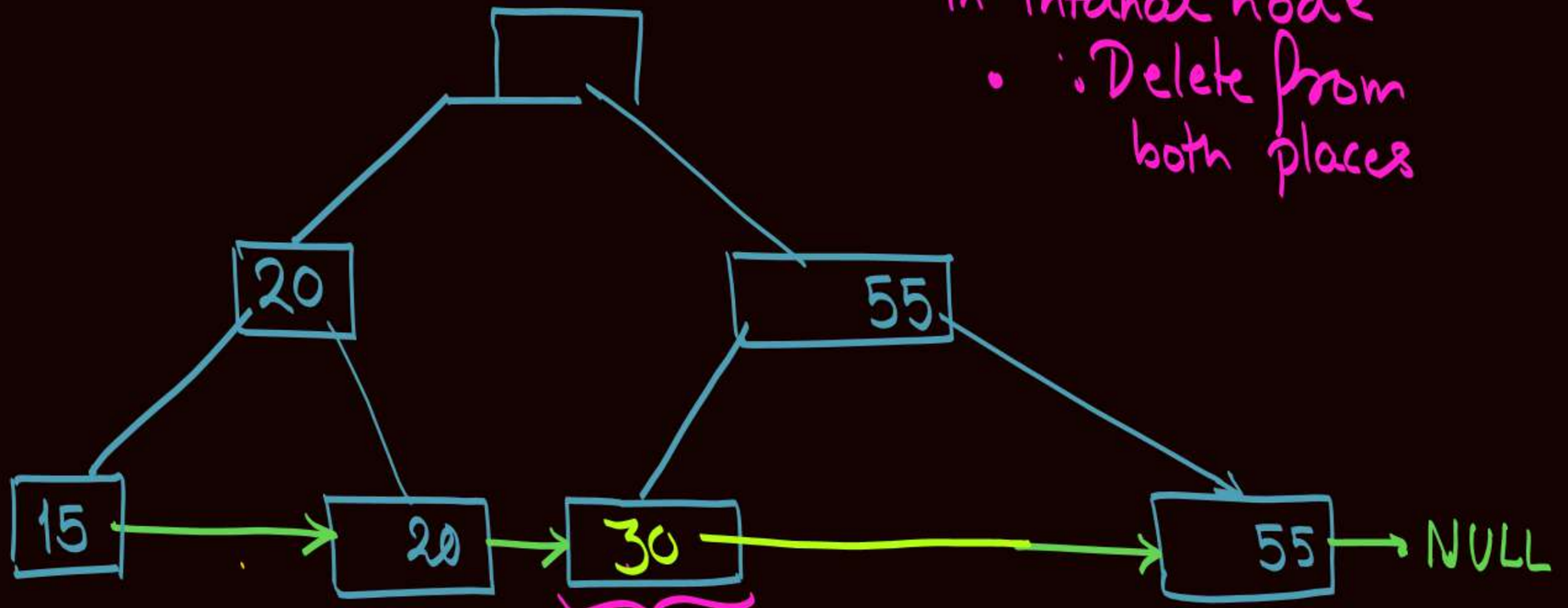
Present in leaf node as well as in internal node
∴ Delete from both places



Underflow
&
None of the sibling can help
∴ Merge with one sibling

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

Present in leaf node as well as in internal node
∴ Delete from both places



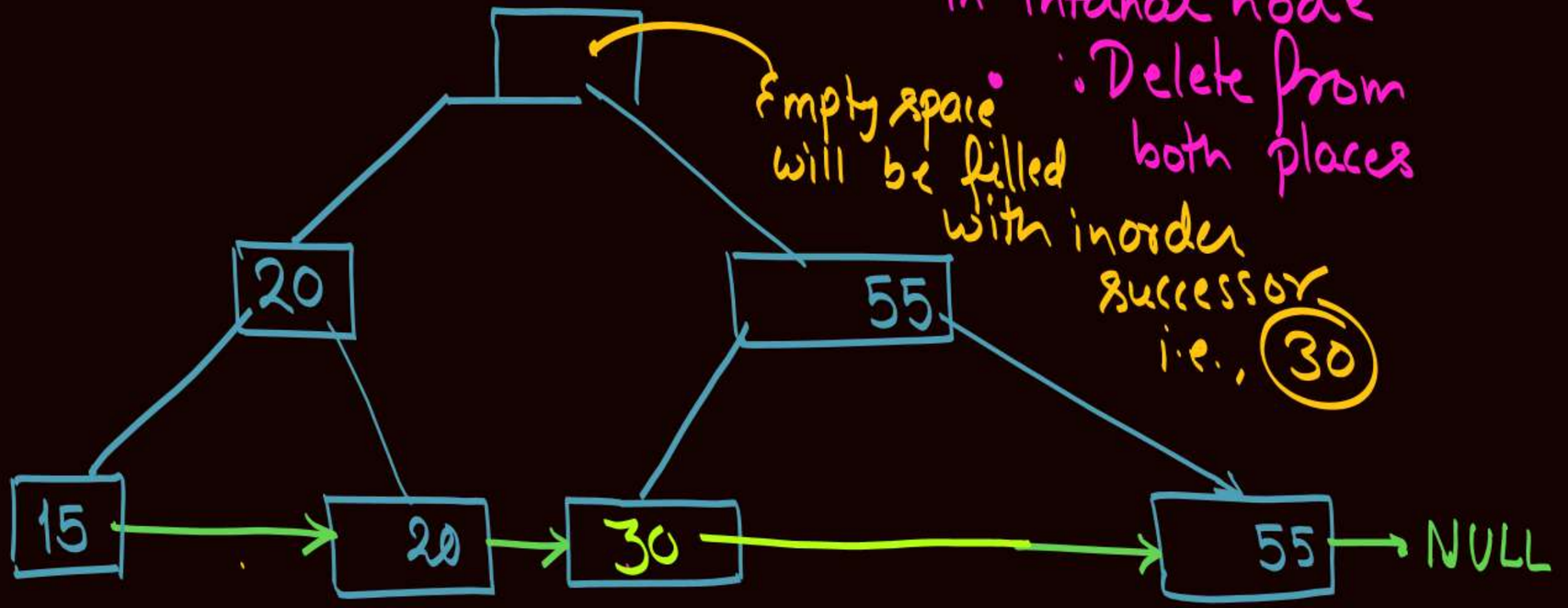
Underflow
&
None of the sibling can help
∴ Merge with one sibling

{ Because of Merging there is no underflow in internal node }

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, 55

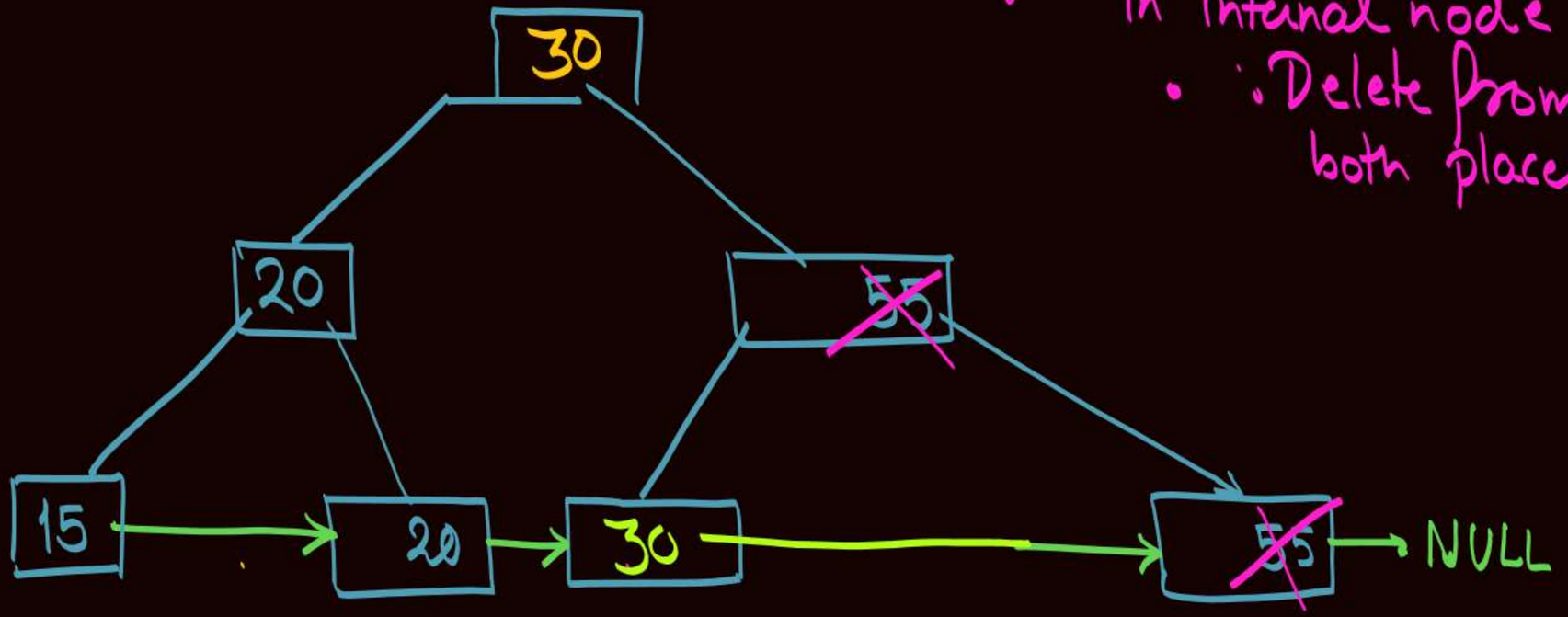
Present in leaf node as well as in internal node

: Delete from both places



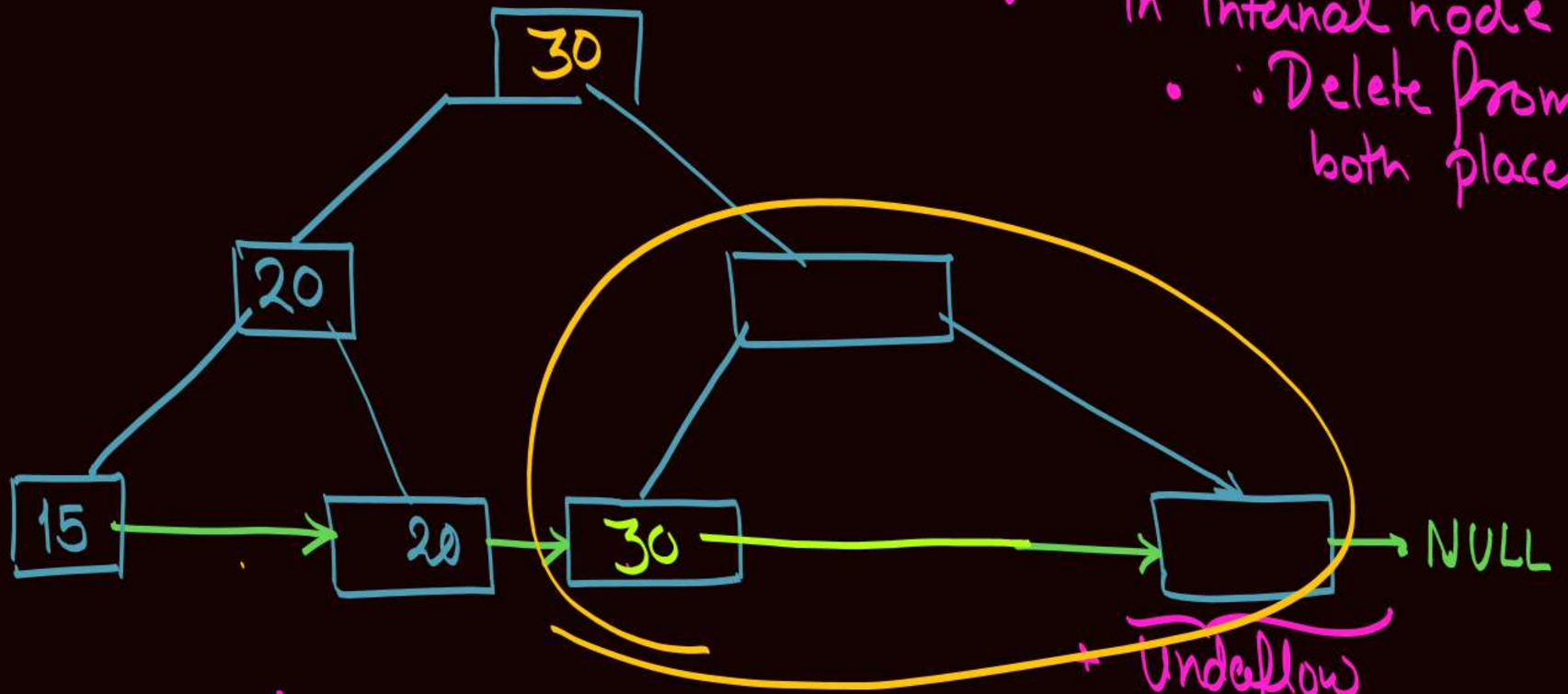
delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~

↳ Present in leaf node as well as in internal node
∴ Delete from both places



delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~

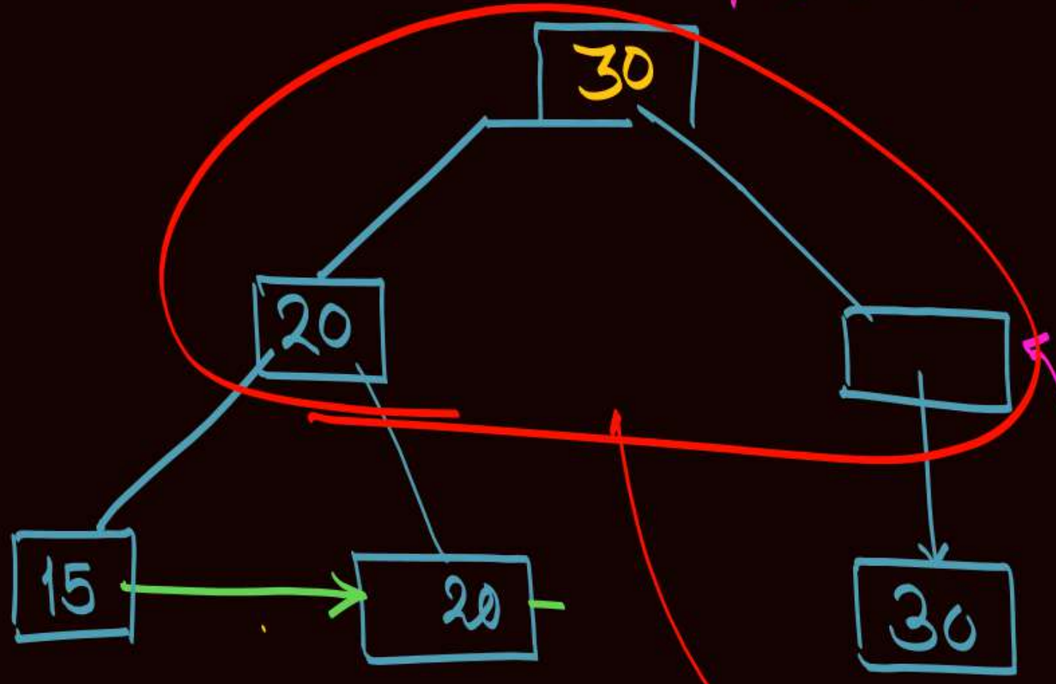
↳ Present in leaf node as well as in internal node
∴ Delete from both places



- Underflow
- Sibling can not help
∴ Merge with sibling

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~

↳ Present in leaf node as well as in internal node
∴ Delete from both places



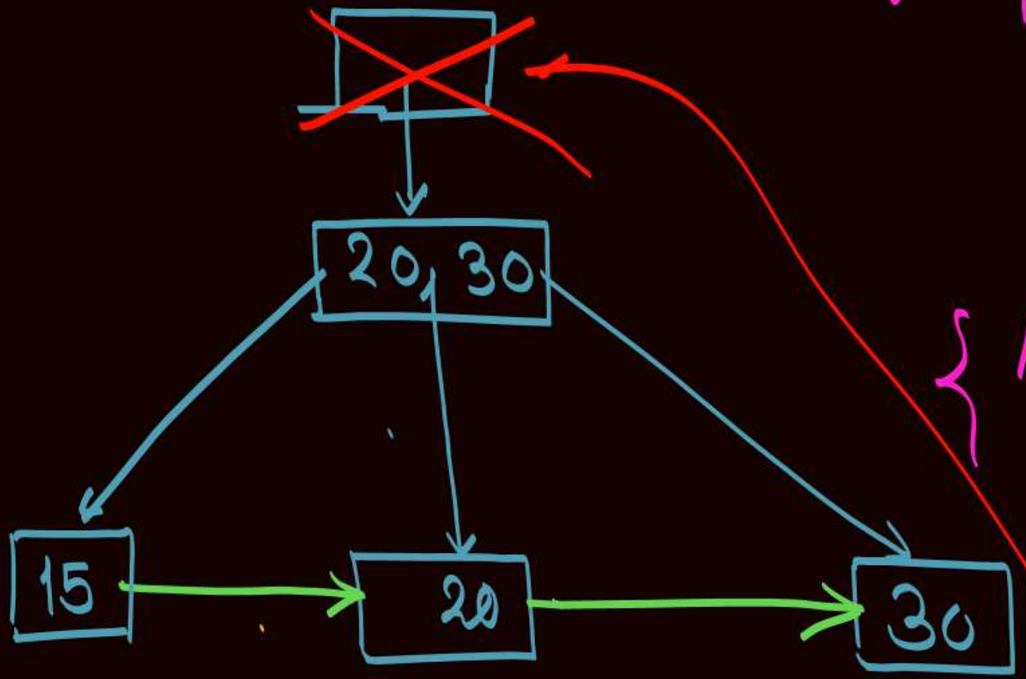
{ Because of merging, there is underflow in internal node

- Sibling of internal node can not help

∴ Merge with sibling

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~

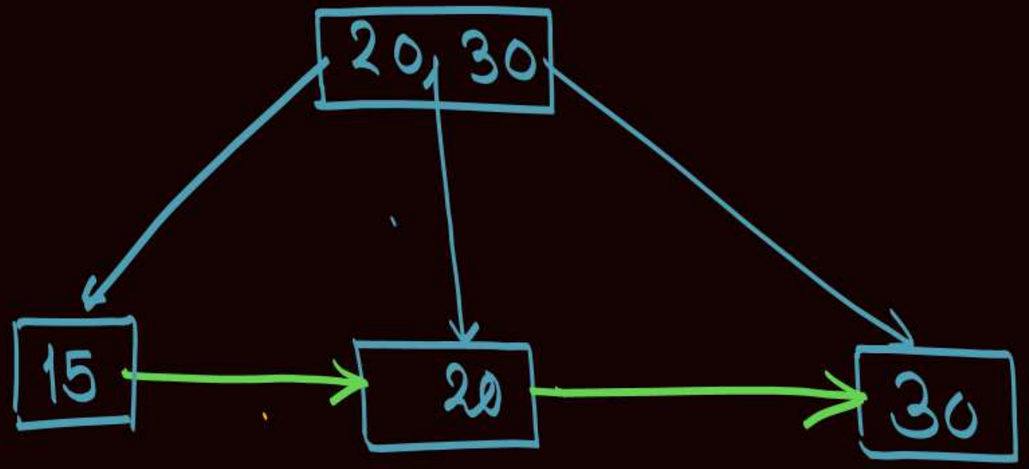
↳ Present in leaf node as well as in internal node
∴ Delete from both places



{ As a result of merging, Underflow in root node,
∴ height of tree will decrease by 1
{ root node with underflow is deleted }

delete ⇒ ~~40~~, ~~5~~, ~~45~~, ~~35~~, ~~25~~, ~~55~~

Final tree after all deletions,



ER Model
&
Integrity Constraints



Topic : Entity-Relationship Model (ER-Model)

- ER Model is a high-level data model diagram which defines the conceptual view of the database (OR) that gives the graphical representation of the logical structure of the database.

- Entity-Relationship model is based on objects called entities, and relationship among these entities.



Topic : Entity

- An entity refers to any object having either a physical existence or a conceptual existence. For example, a school, a person, a house, a university, a company or a job.



Topic : Entity Set

- An entity set is a collection or set of all entities of a particular entity type at any point in time.

"Student"

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19

This complete table is referred as "Student Entity Set" and each row represents an "entity".



Topic : Entity Set

An entity set is represented by a rectangle in ER diagram

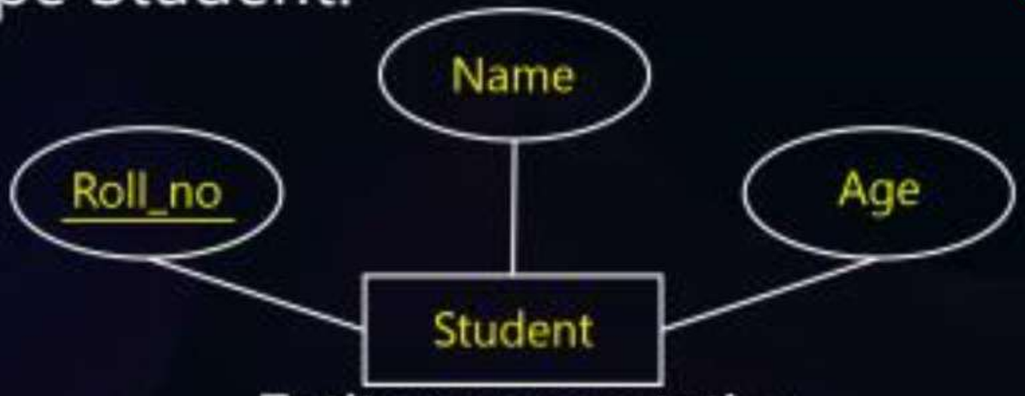
Student



Topic : Attribute

- Each entity is identified uniquely by the values of its attributes.
- An attribute in an Entity-Relationship Model describes the properties or characteristics of an entity. It is represented by an oval or ellipse shape in the ER diagram. Every oval shape represents one attribute and is directly connected to its entity set which is rectangle in shape.

For example, roll_no, name, and age are the attributes which define entity type Student.



Entity representation

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19



Topic : Types of Attributes

i. Simple attribute

ii. Composite attribute

iii. Single-valued attribute

iv. Multi-valued attribute

v. Key attribute

vi. Derived attribute

vii. Partial key attribute



Topic : Types of Attributes

(i) **Simple Attributes:** An attribute which contains an atomic value and cannot be divided further is called a simple attribute. Example, Roll_no, Sex, etc.

(ii) **Composite attribute:** An attribute which can divided further. It is an attribute which is composed of several components. Example name, address, etc.

(iii) **Single-valued attribute:** An attribute which can have only one value. eg Sname

(iv) **Multi-valued attribute :** An attribute which has multiple values (more than one value) eg Mob_no.

(v) **Key attribute :** An attribute which can identify all entities uniquely.

(vi) **Derived attribute :** An attribute whose value can be derived or found from any other attribute is known as derived attributes.

(vii) **Partial key attribute :** An attribute which can identify a group of tuples.



Topic : Types of Attributes



Attribute



Multivalued Attribute



Composite Attribute

Composite attribute



Key Attribute



Partial Attribute



Derived Attribute

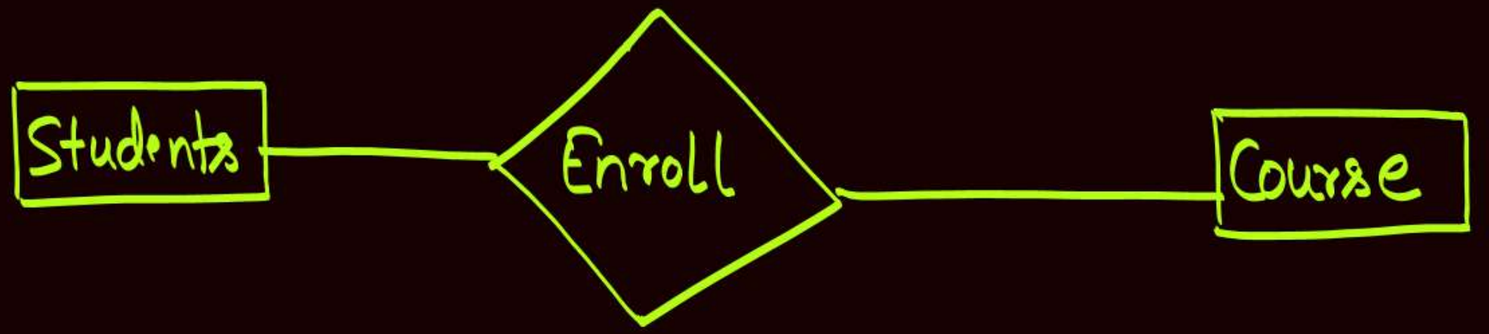
it is associated with weak entity set.



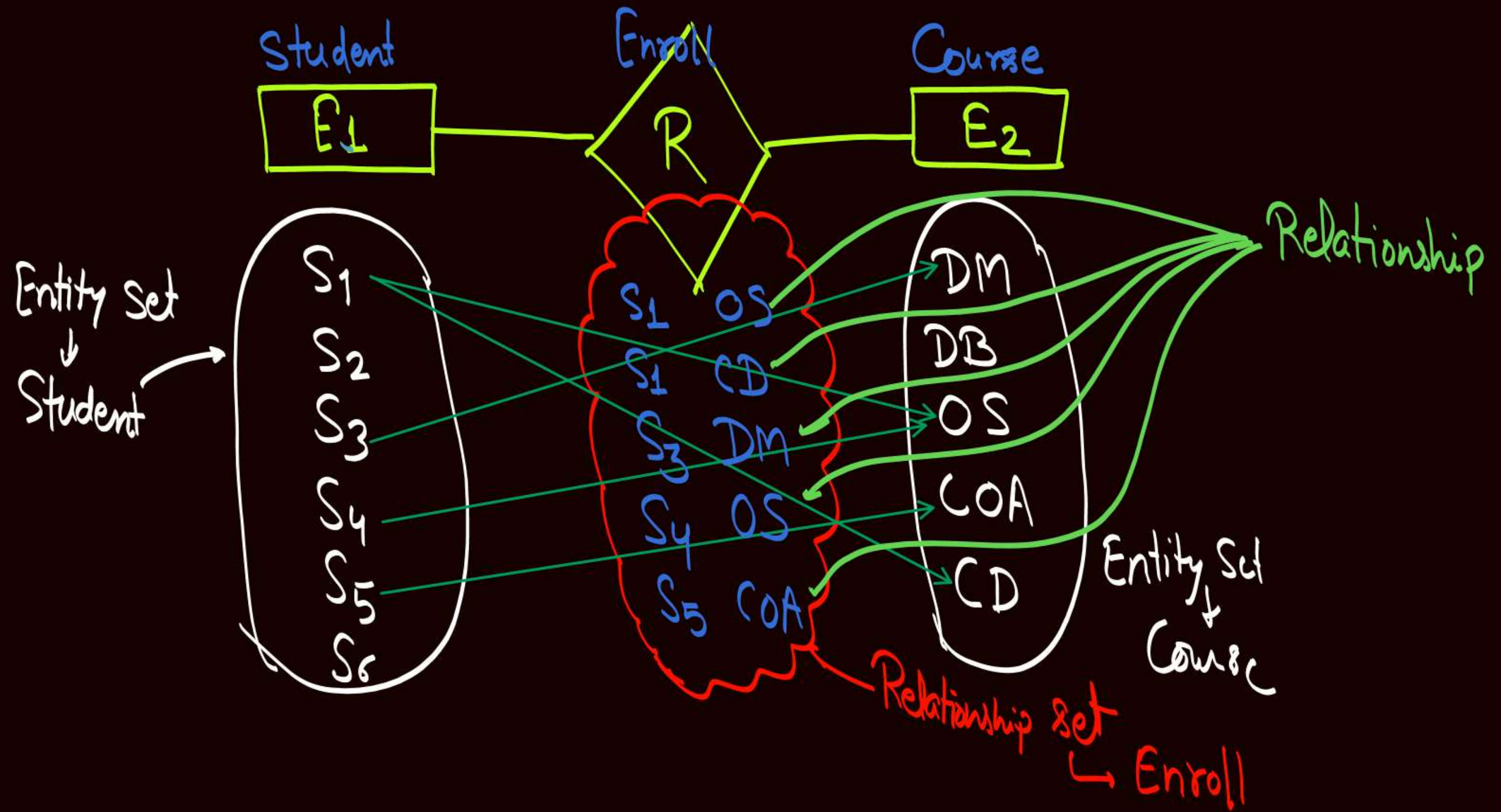
Topic : Relationship in ER model

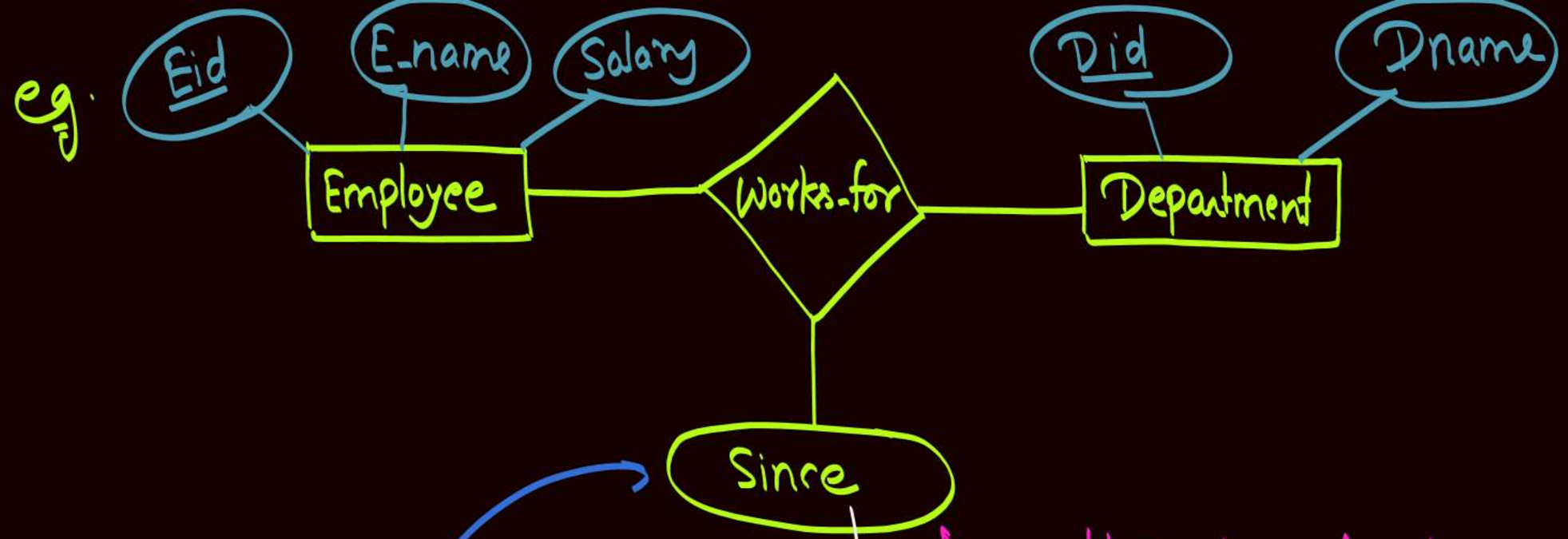
A Relationship Type represents the association between entity types. For example, 'Enrolled' is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.

eg.



eg.


Student
Enroll
Course
E1
R
E2
Entity Set
↓
Student
S1
S2
S3
S4
S5
S6
S1 OS
S1 CD
S3 DM
S4 OS
S5 COA
DM
DB
OS
COA
CD
Relationship
Entity Set
↓
Course
Relationship Set
↳ Enroll



Attributes of the relation are called "descriptive" attributes

attribute of the relation

it will specify that since when the Employee works_for a specific department

Note:

- It is not necessary for all the relationship type to have their own attributes



Topic : Descriptive attribute

Descriptive attributes are use to provide the information related to relationship set.



Topic : Degree of a Relationship

- The number of different entity sets participating in a relationship is called the degree of a relationship.

Degree of a relationship = Number of entity sets participating in a relationship



Topic : Types of Relationship

On the basis of degree of a relationship set, a relationship set can be classified into the following types-

i. Unary Relationship: Unary relationship set is a relationship set where only one entity set participates in a relationship set.

ii. Binary Relationship: Binary relationship set is a relationship set where two entity sets participate in a relationship set.

iii. Ternary Relationship: Ternary relationship set is a relationship set where three entity sets participate in a relationship set.

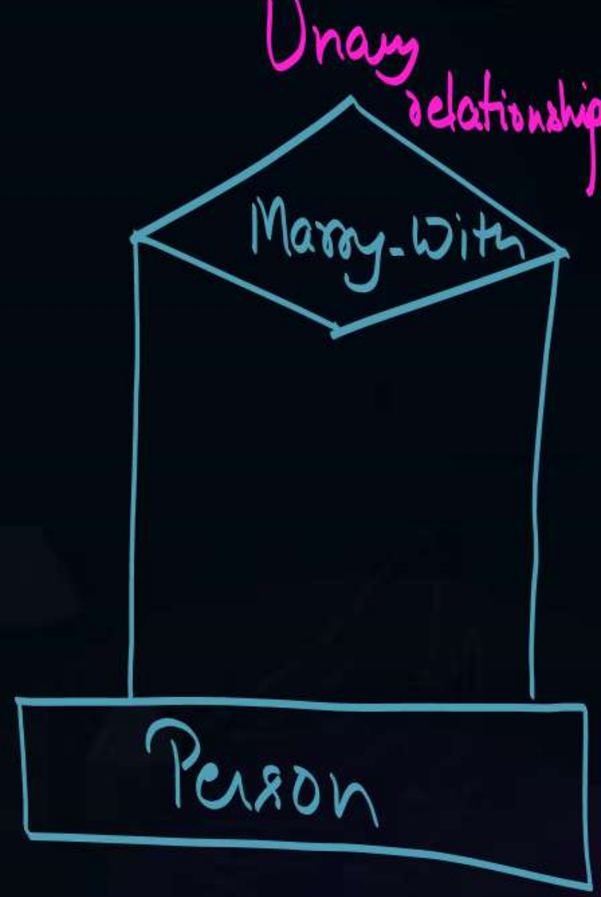
iv. N-ary Relationship: N-ary relationship set is a relationship set where 'N' entity sets participate in a relationship set.



Topic : Unary Relationship



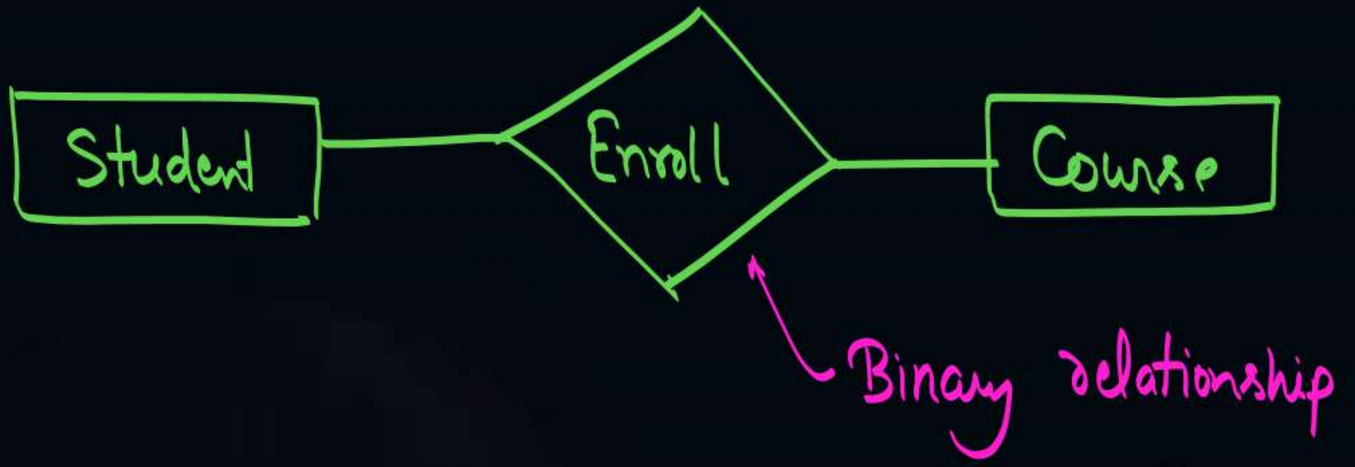
≡



Only one entity set i.e. 'Person'
is participating in relationship set "Marry With"

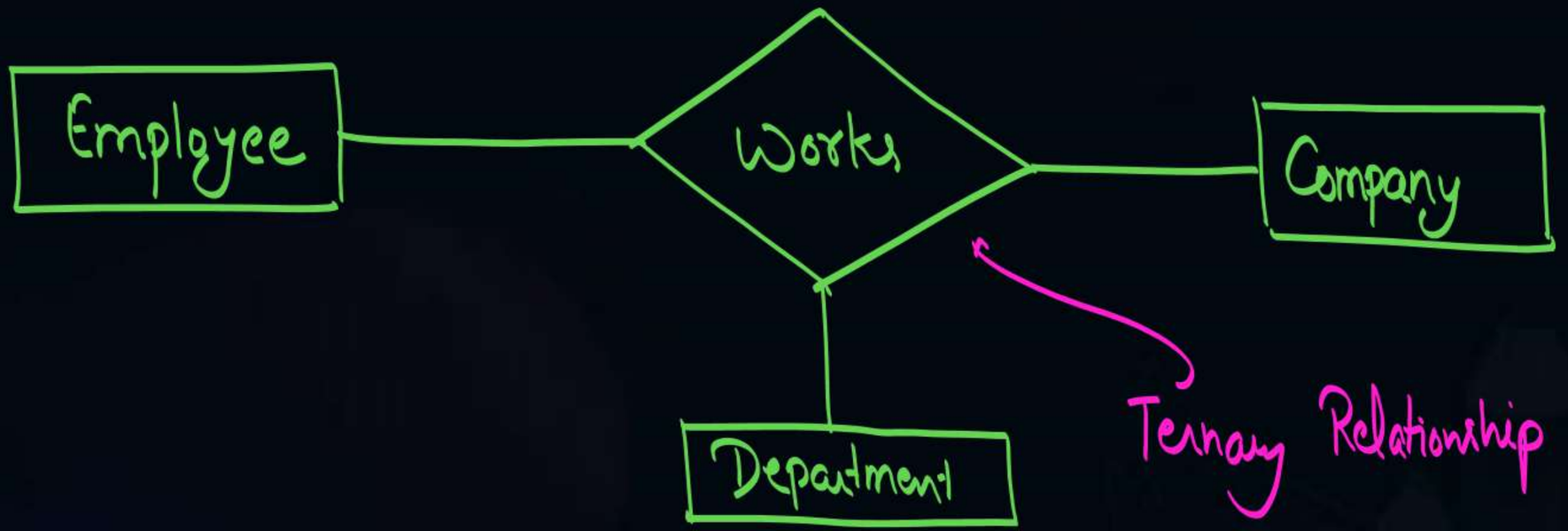


Topic : Binary Relationship



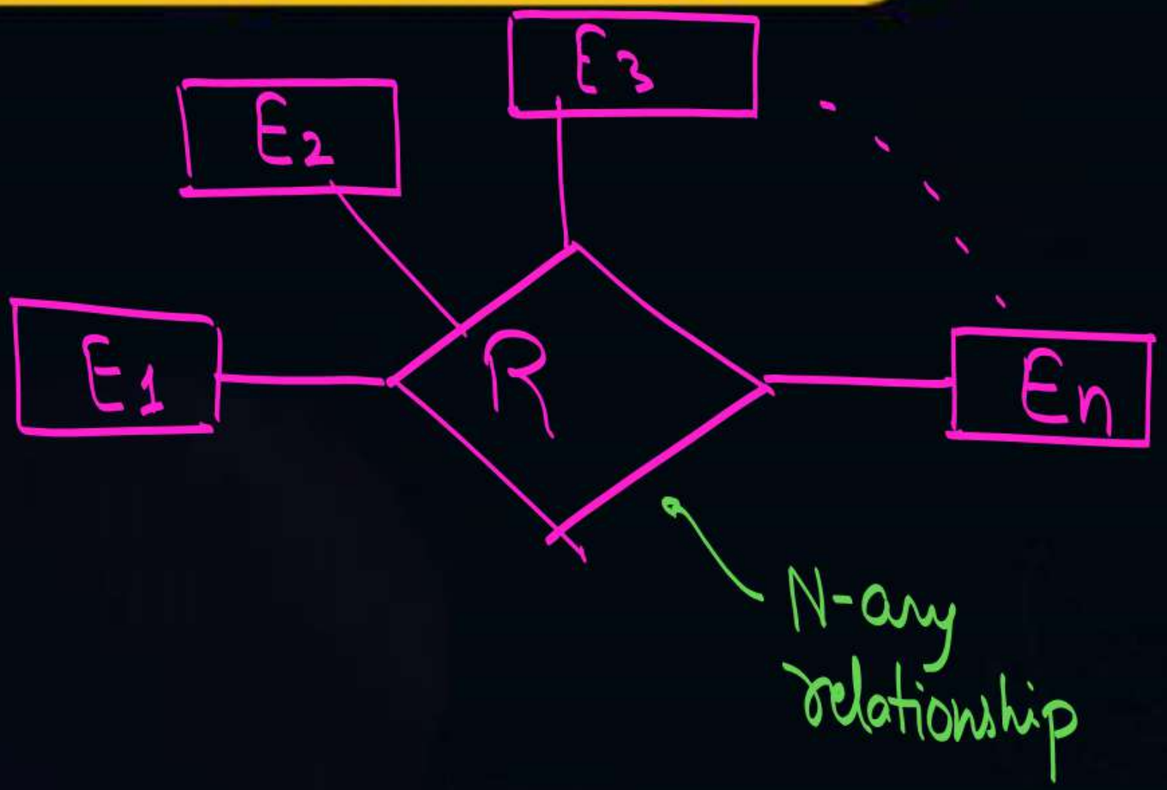


Topic : Ternary Relationship





Topic : N-ary Relationship





Topic : Participation constraints

There are two types of participation constraints :

1. Total participation
2. Partial participation



Topic : Total participation

- If every entity of an entity set participate in the relationship set, then that entity set is said to have total participation.
- Total participation of an entity set in a relationship set is denoted by double line



Topic : Partial participation

- If at least one entity of entity set is not participating in the relationship set, then that entity set is said to have partial participation.
- Partial participation of an entity set in a relationship set is denoted by single line.



Topic : Mapping cardinality

Mapping cardinality or cardinality ratio is used to denote the number of entities to which another entity can be associated through a certain relation set.

For a binary relationship test mapping cardinalities must be one of the following types:

1. One-to-one
2. One-to-many
3. Many-to-one
4. Many-to many



Topic : Strong entity Set

- A strong entity set is an entity set that contains sufficient attributes to uniquely identify all its entities. In other words, a primary key exists for a strong entity set.
- Primary key of a strong entity set is represented by underlining it.



Topic : Weak entity Set

- A weak entity set is an entity set that does not contain sufficient attributes to uniquely identify its entities.
- In other words, a primary key does not exist for a weak entity set.
- However, it contains a partial key called as a discriminator.
- Discriminator can identify a group of entities from the entity set.
- Discriminator is represented by underlining with a dashed line.
- A weak entity must participate in an identifying relationship type with an owner or identifying entity type



Topic : Note

- A weak entity can be identified uniquely only by combining partial key of weak entity set with primary key of owner entity.
- Owner entity set and weak entity set must participate in an one-to-many relationship set. (Owner side is 1 and weak entity side is many)
- Weak entity set must have total participation in identifying relationship set.

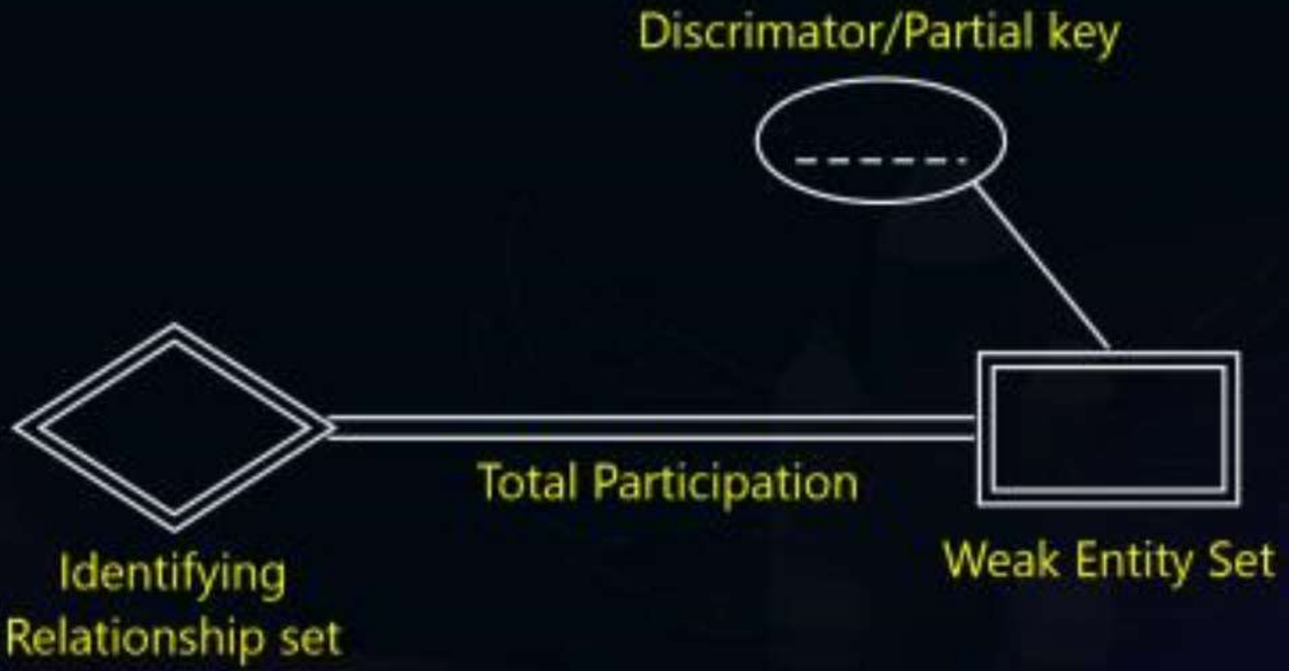


Topic : Differences between Strong entity set and Weak entity set

Strong entity set	Weak entity set
A single line rectangle is used for the representation of a strong entity set.	A double line rectangle is used for the representation of a weak entity set.
It contains sufficient attributes to form its primary key.	It does not contain sufficient attributes to form its primary key.
A single line diamond symbol is used for the representation of the relationship that exists between the two strong entity sets.	A double line diamond symbol is used for the representation of the identifying relationship that exists between the strong and weak entity set.
Total participation may or may not exist in the relationship.	Total participation always exists in the identifying relationship.



Topic : Weak entity set in ER diagram





Topic : Relational Model concept

- The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- The relational model is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Model concept

- The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- The relational is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Constraints

These are the restrictions or set of rules imposed on the database content. It validates the quality of the database. It validates that various operations like data insertion, updation, etc., can be performed without affecting the integrity of the data.

Constraints could be either on a column level or a table level. The column level constraints are applied only to one column, whereas the table level constraints are applied to the whole table.



Topic : Types of Constraints on Relational Model

1. Domain constraint
2. Key constraint or Tuple Uniqueness constraint
3. Entity Integrity constraint
4. Referential Integrity constraint



Topic : Domain constraint

Every domain must contain atomic values(smallest indivisible units) which means composite and multi-valued attributes are not allowed.

We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain. Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype



Topic : Key constraint or Tuple Uniqueness constraint

These are called uniqueness constraints since it ensures that every tuple in the relation should be unique.

A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys.



Topic : Entity Integrity constraint

Entity integrity constraint specifies that no attribute of primary key must contain a null value in any relation.



Topic : Referential Integrity constraint

The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.



Topic : Referential Integrity constraint

Foreign Key: A foreign key is a set of attributes in a table that refers to the primary key or alternate key of same table or some other table.



Topic : Referential Integrity constraint

Referential integrity constraint is enforced through a foreign key

Let foreign key in relation R1 refers to primary key of relation R2.

The values of the foreign key in a tuple of relation R1 can either take the values of the primary key for some tuple in relation R2, or can take NULL values, but can’t be empty.



Topic : Referential Integrity constraints on Referenced relation

Insertion: Insertion of a tuple in referenced relation does not result in any referential integrity violation.

Deletion: Deletion of a tuple from referenced relation may result in referential integrity violation.

Updation: Updation of a tuple in referenced relation may result in referential integrity violation.



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Deletion of a primary key value from the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

On Delete No Action:

On Delete Cascade:

On Delete Set NULL:



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Updation in the primary key value in the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

On Update No Action:

On Update Cascade:

On Update Set NULL:



Topic : Referential Integrity constraints on Referencing relation

Insertion: Insertion of a tuple in referencing relation may result in referential integrity violation.

Deletion: Deletion of a tuple from referencing relation does not result in any referential integrity violation.

Updation: Updation of a tuple in referencing relation may result in referential integrity violation.



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If any operation in referencing relation (child relation) causes any referential integrity violation, then corresponding operation is prohibited.



2 mins Summary

Topic ER model & ER diagram

Topic Relational model & Integrity constraints

THANK - YOU